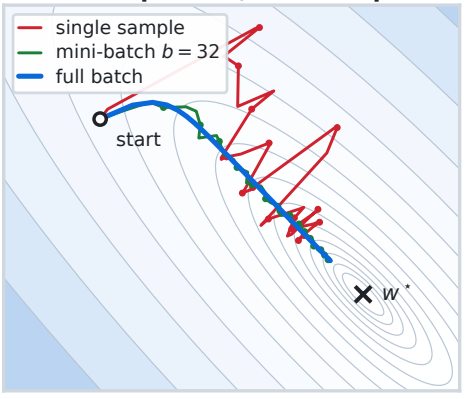
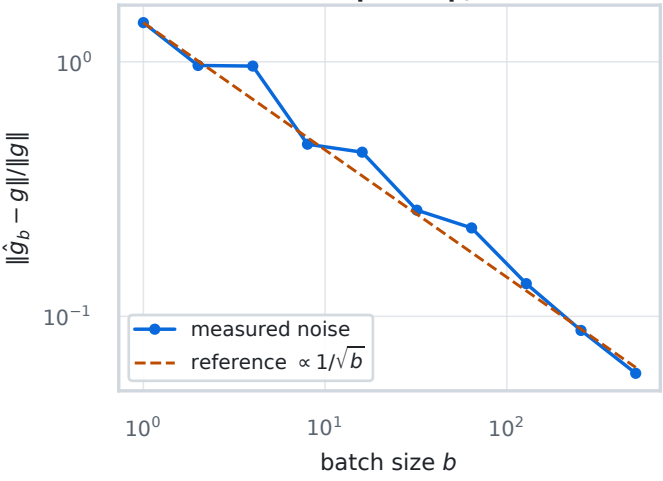

(a) 45 steps each, same step size
single sample
mini-batch $b = 32$
full batch
start
$w^\star$


(b) more data per step, less noise
$\|\hat{g}_b - g\|/\|g\|$
$10^0$
$10^{-1}$
measured noise
reference $\propto 1/\sqrt{b}$
$10^0$
$10^1$
$10^2$
batch size $b$